

Chapter 9 Review

Chapter Summary

After reading this chapter and completing the exercises, you should understand the following about networks.

Identify the basic network architectures

- The two broad categories of networks are LANs and WANs.
- The physical arrangement of a network is typically called the network's topology.
- The four main types of network topologies are ring, bus, star, and mixed.
- Additional architectures include wireless and ad hoc.

Examine segregation, segmentation, and isolation with respect to network architectures

- Understand both physical and logical structures, and their differences.
- The use of airgaps, zones, and conduits in control system architectures is a key element of their network security architecture.
- Enclaves act as local segregation entities.
- Using zero trust for the authentication model changes the old perimeter security model of networks.

Classify security zones

- A DMZ is a buffer zone between networks with different trust levels. Companies often place public resources in a DMZ so that Internet users and internal users may access those resources without exposing the internal company network to the Internet.
- An intranet is a private, internal network that uses common network technologies (HTTP, FTP, and so on) to share information and provide resources to organizational users.
- An extranet is a semiprivate network that uses common network technologies (HTTP, FTP, and so on) to share information and provide resources to business partners.
- An enclave is a specialized security zone with common security requirements.
- A VLAN (or virtual LAN) is a group of ports on a switch configured to create a logical network of computers that appear to be connected to the same network, even if they are located on different physical network segments. Systems on a VLAN can communicate with each other but cannot communicate directly with systems on other VLANs.
- Trunking is the process of spanning a single VLAN across multiple switches.

Explain routing and address translation

- The process of moving packets from one end device to another through different networks is called routing.
- Subnetting is the process of dividing a network address space into smaller networks.
- DHCP allows network devices to be automatically configured on a network and temporarily assigned an IP address.
- Network Address Translation (NAT) converts private, internal IP addresses to public, routable IP addresses, and vice versa.

Examine the Internet Protocol in depth

- IP packets provide the method of moving data, with headers defining the type of packet.
- Different packet types, TCP and UDP, have different levels of overhead and delivery guarantees.
- ICMP packets manage errors and diagnostic issues.
- IPv6 is a complete overhaul of the IP protocol, not just an expanded address space.

Define the basic network protocols

- Protocols, agreed-upon formats for exchanging or transmitting data between systems, enable computers to communicate.
- When data is transmitted over a network, it is usually broken up into smaller pieces called packets.
- Most protocols define the types and format for the packets used in them.
- TCP is connection oriented, requires the three-way handshake to initiate a connection, and provides guaranteed and reliable data delivery.
- UDP is connectionless, lightweight, and provides limited error checking and no delivery guarantee.
- Each network device has a unique hardware address known as a MAC address. The MAC address is used for packet delivery.
- Network devices are also typically assigned a 32-bit number known as an IP address.
- The Domain Name Service (DNS) translates names like www.cnn.com into IP addresses.

Explore additional elements of networking

- Internetworking covers how different networks are connected using BGP and other protocols.
- MPLS covers how packets can traverse telecom networks with switching to guarantee performance.
- Software-defined networks (SDNs) separate the data and management planes to gain flexibility and capability.
- Quality of service (QoS) recognizes that not all packets need the same level of technical timing and allow for flexibility in capacity.
- Traffic engineering comprises the steps to achieve quality of service and performance metrics.
- Route security examines the concepts of end-to-end protection of communications.

Key Terms

ad hoc (272)

Address Resolution Protocol (ARP) (294)

airgap (275)

autonomous systems (ASs) (303)

Border Gateway Protocol (BGP) (303)

bus topology (270)

- control systems (276)
- datagram (284)
- denial of service (DOS) (288)
- DMZ (277)
- DNS over HTTPS (DoH) (297)
- DNSSEC (296)
- Domain Name System (DNS) (295)
- Dynamic Host Configuration Protocol (DHCP) (298)
- east-west traffic (279)
- enclave (273)
- Ethernet (293)
- extranet (280)
- flat network (281)
- guest (280)
- hardened operating system (277)
- honeynet (280)
- honeypot (280)
- Internet Control Message Protocol (ICMP) (287)
- Internet Protocol (IP) (283)
- intranet (279)
- local area network (LAN) (269)
- maximum transmission unit (MTU) (283)
- Media Access Control (MAC) address (293)
- mixed topology (270)
- multi-protocol label switching (MPLS) (303)
- network (268)
- Network Address Translation (NAT) (301)
- network segmentation (272)
- packet (283)
- packet fragmentation (283)
- physical separation (273)
- protocol (281)
- quality of service (QoS) (305)
- ring topology (270)
- routing (295)
- software-defined networking (303)
- star topology (270)
- storage area network (SAN) (269)
- subnet mask (299)
- subnetting (299)
- three-way handshake (286)
- topology (270)
- Transmission Control Protocol (TCP) (285)
- trunking (274)
- User Datagram Protocol (UDP) (285)
- virtual local area network (VLAN) (270)

virtualization (275)
wide area network (WAN) (269)
wireless (271)
zero trust (276)
zones (276)

Key Terms Quiz

Use terms from the Key Terms list to complete the sentences that follow. Don't use the same term more than once. Not all terms will be used.

1. A(n) _____ is a group of two or more devices linked together to share data.
2. A packet in an IP network is sometimes called a(n) _____.
3. Moving packets from source to destination across multiple networks is called _____.
4. The _____ is the hardware address used to uniquely identify each device on a network.
5. A(n) _____ tells you what portion of a 32-bit IP address is being used as the network ID and what portion is being used as the host ID.
6. The shape or arrangement of a network, such as bus, star, ring, or mixed, is known as the _____ of the network.
7. A small, typically local network covering a relatively small area such as a single floor of an office building is called a(n) _____.
8. A(n) _____ is an agreed-upon format for exchanging information between systems.
9. The packet exchange sequence (SYN, SYN/ACK, ACK) that initiates a TCP connection is called the _____.
10. _____ is the protocol that allows the use of private, internal IP addresses for internal traffic and public IP addresses for external traffic.

Multiple-Choice Quiz

1. What is Layer 1 of the OSI model called?
 - A. The physical layer
 - B. The network layer
 - C. The initial layer
 - D. The presentation layer

2. Which of the following statements best describes the User Datagram Protocol (UDP)?
 - A. It provides excellent error-checking algorithms.
 - B. It is a connectionless protocol.
 - C. It guarantees delivery of packets.
 - D. It requires a permanent connection between the source and destination.
3. What is the process that dynamically assigns an IP address to a network device called?
 - A. NAT
 - B. DNS
 - C. DHCP
 - D. Routing
4. What is the three-way handshake sequence used to initiate TCP connections?
 - A. ACK, SYN/ACK, ACK
 - B. SYN, SYN/ACK, ACK
 - C. SYN, SYN, ACK/ACK
 - D. ACK, SYN/ACK, SYN
5. Which of the following is a control and information protocol used by network devices to determine such things as a remote network's availability and the length of time required to reach a remote network?
 - A. UDP
 - B. NAT
 - C. TCP
 - D. ICMP
6. What is the name of the protocol that translates names into IP addresses?
 - A. TCP
 - B. DNS
 - C. ICMP
 - D. DHCP
7. Dividing a network address space into smaller, separate networks is called what?
 - A. Translating
 - B. Network configuration
 - C. Subnetting
 - D. Address translation
8. Which protocol translates private (nonroutable) IP addresses into public (routable) IP addresses?
 - A. NAT
 - B. DHCP
 - C. DNS
 - D. ICMP

9. Which of the following statements best describes the Transmission Control Protocol (TCP)?
- A. It is connectionless.
 - B. It provides no error checking.
 - C. It allows for packets to be processed in the order they were sent.
 - D. It has no overhead.
10. Which of the following would be a valid MAC address?
- A. 00:07:e9
 - B. 00:07:e9:7c:c8
 - C. 00:07:e9:7c:c8:aa
 - D. 00:07:e9:7c:c8:aa:ba

Essay Quiz

1. A developer in your company is building a new application and has asked you if she should use TCP- or UDP-based communications. Provide her with a brief discussion of the advantages and disadvantages of each protocol.
2. Your boss wants to know if DHCP is appropriate for both server and PC environments. Provide her with your opinion and be sure to include a discussion of how DHCP works.
3. Describe three basic types of network topologies and provide a sample diagram of each type.
4. Describe the three-way handshake process used to initiate TCP connections.
5. Your boss wants to know how subnetting works. Provide her with a brief description and be sure to include an example to illustrate how subnetting works.

Lab Projects

Lab Project 9.1

A client of yours only has five external, routable IP addresses but has over 50 systems that they want to be able to reach the Internet for web surfing, e-mail, and so on. Design a network solution for the client that addresses their immediate needs but will still let them grow in the future.

Lab Project 9.2

Your boss wants you to learn how to use the arp and nslookup commands. Find a Windows system and open a command/DOS prompt. Type in arp and press ENTER to see the options for the arp command. Use the arp command to find the MAC address of your system and at least five other systems on your network. When you are finished with arp, type in nslookup and press ENTER. At the prompt, type in the name of your favorite website, such as www.cnn.com. The nslookup command will return the IP addresses that match that domain name. Find the IP addresses of at least five different websites.

Conklin, Wm. Arthur; White, Greg; Cothren, Chuck; Davis, Roger L.; Williams, Dwayne.
Principles of Computer Security: CompTIA Security+ and Beyond, Sixth Edition (Exam SY0-601) (pp. 306-309). McGraw Hill LLC. Kindle Edition.